

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460504.7201 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.054
Transit depth (Rp/Rs)^2	3.44 +/- 2.01 [%]
Orbital Inclination (inc)	87.38 +/- 1.69
Airmass coefficient 1 (a1)	0.308 +/- 0.092
Airmass coefficient 2 (a2)	0.659 +/- 0.099
Scatter in the residuals of the lightcurve fit is	25.44 %
Best Comparison Star	#1 - [453, 290]
Optimal Aperture	24.23
Optimal Annulus	234.21
Transit Duration (day)	0.077 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.185 $\pm$ 0.054 vs 0.1406 (deviation 0.36 σ)
Duration fit vs published	0.077 d vs 0.0975 d (deviation 0.48 σ)
Mid-time O-C	-11.0 min
Depth SNR	1.5
Residual scatter	25.44 %

Light curve

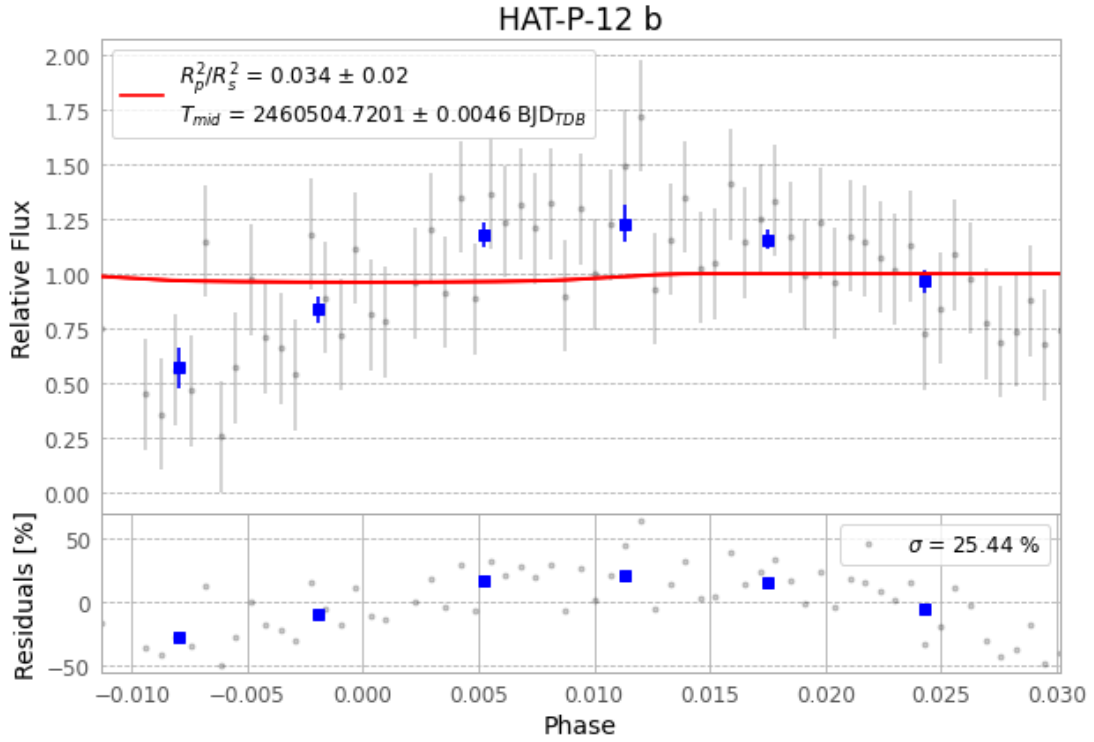


Figure 1 — FinalLightCurve_HAT-P-12 b_2024-07-12.png (SHA-256 a1c2d2a567dcfd56...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [101, 11], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 376f9681091f9449...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

79 FITS frames (52 MB), HATP-12240713034515.FITS → HATP-12240713073915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-240713.log (711bbb92eb00...), FinalLightCurve_HAT-P-12 b_2024-07-12.png (a1c2d2a567dc...), FinalLightCurve_HAT-P-12 b_2024-07-12.csv (82e1bb99edb8...)

Compute resources

Wall time 967.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 00:43Z.